

BONN ROLEBER, GermanyWMO: **105190**Lat: **50.7350N**Lon: **7.1931E**Elev: **160**StdP: **99.41**Time Zone: **1.00 (EUC)**Period: **01-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-5.7	-11.6	1.4	-5.5	-9.4	1.7	-3.8	9.2	8.6	8.3	6.9	2.1	110	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.2	19.8	27.3	19.1	25.6	18.3	20.9	27.0	20.1	25.5	19.2	24.1	3.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.2	22.9	18.1	13.3	21.9	17.4	12.7	21.1	61.2	26.8	58.3	25.9	55.4	24.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.4	5.7	-10.4	33.2	3.2	1.4	-12.7	34.2	-14.5	35.0	-16.3	35.8	-18.6	36.9		
(5)			-10.8	22.5	3.2	0.9	-13.1	23.1	-15.0	23.6	-16.8	24.1	-19.1	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	2.4	2.9	6.3	10.4	13.7	16.8	18.8	18.3	14.9	11.0	6.9	2.8
	DBStd	6.97	4.65	4.40	4.33	4.04	3.65	3.30	3.24	2.77	3.12	3.77	3.46	4.07
	HDD10.0	983	238	201	127	43	9	1	0	0	1	34	103	224
	HDD18.3	2988	495	433	372	238	148	67	34	35	112	227	343	482
	CDD10.0	1154	2	2	13	56	125	205	273	256	148	66	10	1
	CDD18.3	118	0	0	0	1	6	22	49	33	8	1	0	0
	CDH23.3	973	0	0	0	16	51	176	414	249	66	1	0	0
	CDH26.7	235	0	0	0	0	9	32	120	65	8	0	0	0
Wind	WSAvg	2.8	3.2	3.2	3.0	2.8	2.7	2.5	2.5	2.5	2.5	2.8	3.0	3.2
Precipitation	PrecAvg	703	49	47	47	43	62	77	76	76	60	56	55	56
	PrecMax	819	112	109	97	87	116	140	189	156	135	123	110	128
	PrecMin	552	7	8	4	0	21	30	31	22	17	28	0	17
	PrecStd	74	26	23	24	21	26	29	36	42	34	24	21	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.1	19.2	25.5	27.8	30.0	32.5	32.0	27.9	22.1	16.4	12.9
		MCWB	10.7	10.0	11.6	14.7	18.9	20.7	20.5	20.7	18.8	17.1	12.4	10.9
	2%	DB	11.8	12.0	16.6	22.6	24.4	27.4	29.7	28.2	25.2	19.7	14.6	11.0
		MCWB	9.6	9.4	10.8	13.7	16.4	19.5	19.8	19.5	18.4	15.4	11.6	9.4
	5%	DB	10.4	10.6	14.7	20.0	22.5	25.4	27.6	26.1	22.9	17.9	13.3	9.6
		MCWB	8.7	8.1	10.2	12.6	15.3	18.1	19.2	19.1	17.2	14.5	11.0	8.1
	10%	DB	8.8	9.0	12.7	17.7	20.6	23.4	25.6	24.0	20.6	16.4	11.7	8.3
		MCWB	7.3	7.0	9.1	11.8	14.5	17.4	18.6	17.9	15.7	13.7	9.9	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.7	13.0	15.7	20.2	22.0	22.2	21.5	20.2	17.5	13.0	11.2
		MCDB	12.7	12.5	16.4	22.4	26.2	27.9	28.2	27.5	25.7	20.9	15.0	12.5
	2%	WB	9.9	9.6	11.8	14.4	18.2	20.4	21.0	20.5	18.9	16.0	12.0	9.5
		MCDB	11.4	11.6	15.0	20.9	22.5	25.7	27.0	26.5	24.0	18.8	14.0	10.6
	5%	WB	8.9	8.4	10.8	13.4	16.5	19.2	20.1	19.6	17.6	15.0	11.2	8.3
		MCDB	10.3	10.2	13.9	18.7	20.5	23.6	25.5	24.8	21.8	17.5	13.0	9.4
	10%	WB	7.3	7.1	9.5	12.3	15.1	18.1	19.2	18.6	16.5	13.9	10.0	7.0
		MCDB	8.6	8.8	12.3	17.0	19.0	22.0	24.3	22.8	20.0	15.9	11.6	8.3
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	7.4	9.4	9.1	9.5	9.4	9.0	8.6	7.1	5.0	4.2
		MCDBR	5.2	6.5	10.4	13.5	12.7	13.1	13.6	13.0	12.3	9.1	6.0	4.9
		MCWBR	4.3	4.3	5.8	6.5	6.1	6.1	5.5	5.8	6.3	5.3	4.1	4.2
	5% WB	MCDBR	5.4	5.2	8.8	11.6	11.1	11.2	11.8	11.1	11.1	8.2	5.7	4.9
		MCWBR	4.6	4.0	5.1	5.8	5.9	5.9	5.2	5.4	6.0	5.2	4.1	4.4
Clear-Sky Solar Irradiance	taub	0.313	0.338	0.369	0.403	0.409	0.411	0.416	0.410	0.380	0.358	0.334	0.311	
	taud	2.432	2.382	2.312	2.253	2.270	2.279	2.264	2.309	2.373	2.429	2.429	2.414	
	Ebn at Noon	720	790	826	836	848	847	835	821	807	756	681	658	
	Edh at Noon	61	82	104	124	127	127	127	116	98	78	61	54	
All-Sky Solar Radiation	RadAvg	0.75	1.44	2.56	4.10	4.85	5.32	5.03	4.32	3.19	1.86	0.87	0.57	
	RadStd	0.12	0.27	0.39	0.48	0.45	0.50	0.60	0.42	0.40	0.19	0.12	0.08	